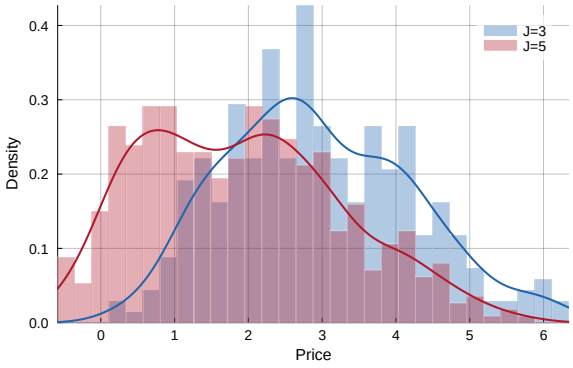
Economics 450-1, Problem Set 1: Figures and Tables

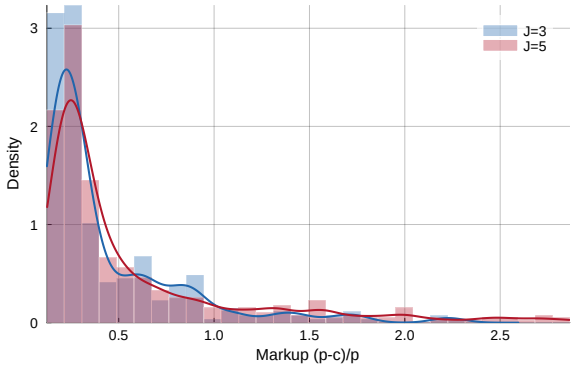
Question 6: Market simulation and welfare

6a

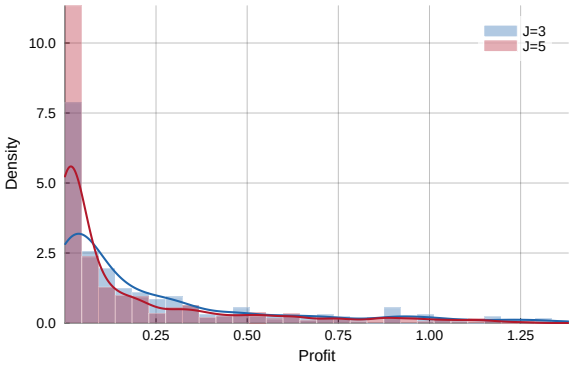
Figure 1: Prices, markups and profits, $J = 3$ versus $J = 5$ (tails not drawn).



(a) Prices



(b) Markups



(c) Profits

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

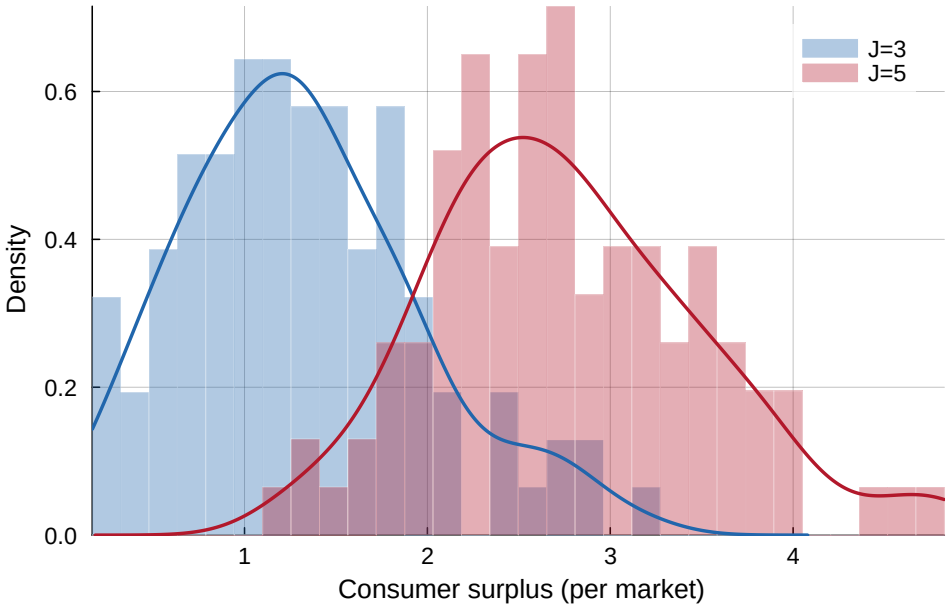
Table 1: Descriptive statistics of prices, markups and profits (all products and markets).

Variable	J	Mean	SD	Median	Q25	Q75
Price	3	3.027	1.371	2.828	1.988	3.964
Price	5	1.891	1.435	1.816	0.780	2.830
Markup	3	0.472	0.634	0.274	0.197	0.574
Markup	5	−0.856	38.116	0.315	0.222	0.782
Profit	3	0.309	0.463	0.105	0.021	0.353
Profit	5	0.254	0.514	0.041	0.004	0.225

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws; full sample, no trimming.

6b

Figure 2: Distribution of consumer surplus across markets, $J = 3$ versus $J = 5$.



Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws; expected surplus averaged over the simulated consumers.

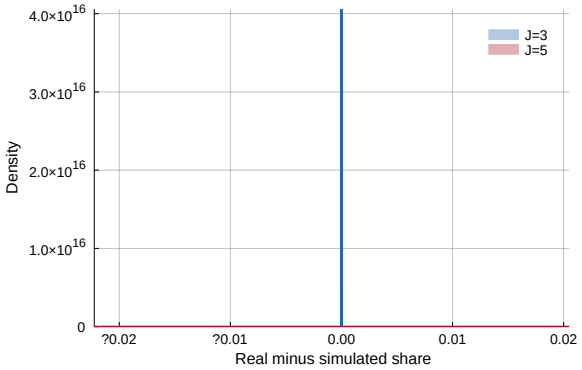
Table 2: Descriptive statistics of consumer surplus across markets.

J	Mean	SD	Median	Q25	Q75
3	1.314	0.642	1.251	0.851	1.729
5	2.765	0.747	2.697	2.258	3.228

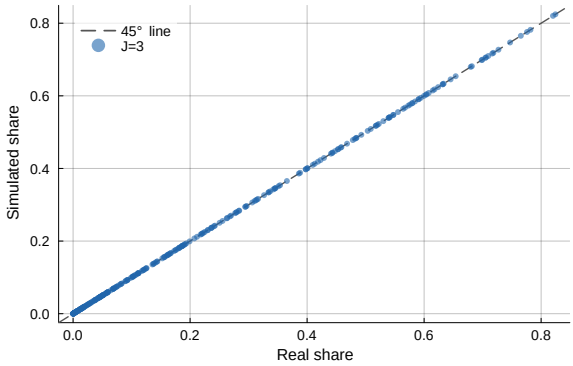
Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

6c

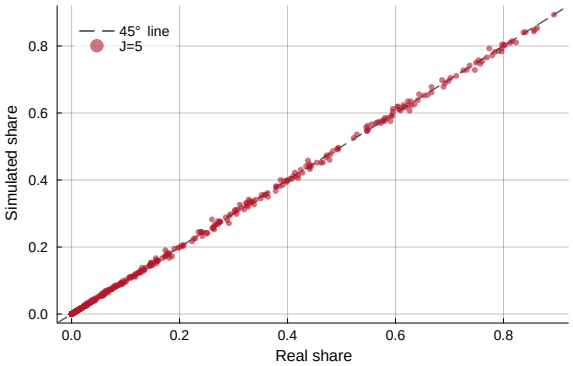
Figure 3: Real versus simulated shares.



(a) Difference, real minus simulated



(b) Real versus simulated, $J = 3$



(c) Real versus simulated, $J = 5$

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Table 3: Discrepancy between real and simulated shares (real minus simulated), and maximum deviation of simulated shares plus outside share from one.

J	Mean diff.	Mean abs. diff.	RMSE	Correlation	Max $\lvert \sum_j s_{jt} + s_{0t} - 1 \rvert$
3	−0.00000	0.00000	0.00000	1.00000	9.99×10^{-16}
5	0.00001	0.00264	0.00481	0.99980	1.33×10^{-15}

Source: true-parameter simulation of `100markets3products.mat` and `100markets5products.mat` with common consumer draws.

Question 7: Demand moments and instruments

7a

Table 4: Sample analogues of the moments $E[\xi_{jt} z_{jt}]$ and t-tests of $H_0 : E[\xi_{jt} z_{jt}] = 0$.

Moment	Mean	SD	t-stat	p-value	N
$E[\xi_{jt} \cdot 1]$	0.0435	1.0037	0.750	0.454	300
$E[\xi_{jt} x_{2jt}]$	0.0207	0.5822	0.617	0.538	300
$E[\xi_{jt} x_{3jt}]$	0.0355	0.9649	0.638	0.524	300
$E[\xi_{jt} p_{jt}]$	0.2950	3.6702	1.392	0.165	300
$E[\xi_{jt} \bar{p}_{j,-t}]$	0.1437	3.1692	0.785	0.433	300

Source: true structural errors ξ_{jt} and data of `100markets3products.mat`.

7c

Table 5: Sample analogues of the moments with the BLP rival-characteristic sums and t-tests of $H_0 : E[\xi_{jt} z_{jt}] = 0$.

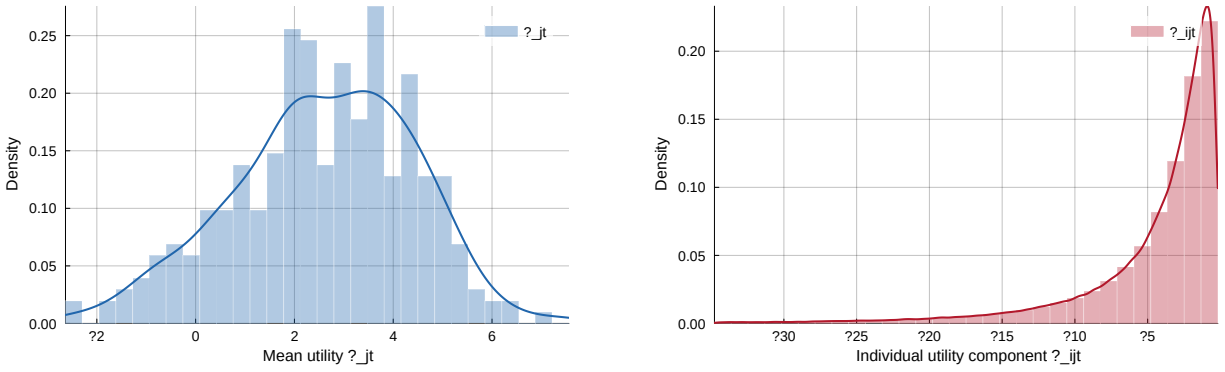
Moment	Mean	SD	t-stat	p-value	N
$E[\xi_{jt} \sum_{k \neq j} x_{2kt}]$	0.0457	1.1542	0.686	0.493	300
$E[\xi_{jt} \sum_{k \neq j} x_{3kt}]$	−0.0087	1.2301	−0.123	0.902	300

Source: true structural errors ξ_{jt} and data of `100markets3products.mat`.

Question 8: Mixed logit by nested fixed point

8a

Figure 4: Mean utility and individual utility component at the true parameters.



(a) Mean utility δ_{jt}

(b) Individual component μ_{ijt} (tails not drawn)

Source: true-parameter simulation of `100markets3products.mat`; $\delta_{jt} = x'_{jt}\beta - \bar{\alpha} p_{jt} + \xi_{jt}$, $\mu_{ijt} = -\sigma_\alpha \nu_i p_{jt}$.

8c

Table 6: Estimates by starting point, T=100, J=3, BLP instruments (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

Start σ_0	Constant	x_2	x_3	Price	σ_α	GMM obj.
0.594	5.589 (4.34×10^7)	1.100 (9.58×10^6)	1.082 (3.21×10^6)	−1.168 (1.41×10^7)	1.054 (3.32×10^6)	5.23×10^{-3}
0.908	5.589 (1.53×10^7)	1.100 (3.36×10^6)	1.082 (1.13×10^6)	−1.168 (4.96×10^6)	1.054 (1.17×10^6)	5.23×10^{-3}
0.859	5.589 (5.26×10^7)	1.100 (1.16×10^7)	1.082 (3.89×10^6)	−1.168 (1.71×10^7)	1.054 (4.02×10^6)	5.23×10^{-3}
1.426	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
0.182	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
0.876	**5.589 (n.a.)**	**1.100 (n.a.)**	**1.082 (n.a.)**	**−1.168 (n.a.)**	**1.054 (n.a.)**	5.23×10^{-3}
0.254	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
1.064	5.589 (1.44×10^7)	1.100 (3.17×10^6)	1.082 (1.06×10^6)	−1.168 (4.67×10^6)	1.054 (1.10×10^6)	5.23×10^{-3}
0.228	5.589 (2.56×10^7)	1.100 (5.64×10^6)	1.082 (1.89×10^6)	−1.168 (8.31×10^6)	1.054 (1.96×10^6)	5.23×10^{-3}
1.168	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
0.768	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
1.378	5.589 (2.19×10^7)	1.100 (4.82×10^6)	1.082 (1.62×10^6)	−1.168 (7.10×10^6)	1.054 (1.67×10^6)	5.23×10^{-3}
1.088	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
0.770	5.589 (1.14×10^7)	1.100 (2.52×10^6)	1.082 (8.47×10^5)	−1.168 (3.72×10^6)	1.054 (8.75×10^5)	5.23×10^{-3}
0.607	5.589 (n.a.)	1.100 (n.a.)	1.082 (n.a.)	−1.168 (n.a.)	1.054 (n.a.)	5.23×10^{-3}
0.444	5.589 (1.35×10^7)	1.100 (2.99×10^6)	1.082 (1.00×10^6)	−1.168 (4.40×10^6)	1.054 (1.04×10^6)	5.23×10^{-3}
True	5.000	1.000	1.000	−1.000	1.000	

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` (per-start values recomputed at each stored σ minimizer, no re-optimization). Bold: start with the lowest GMM objective across starts. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

Table 7: Convergence criterion and contraction error by starting point, T=100, J=3.

Start σ_0	$\hat{\sigma}_\alpha$	GMM obj.	Converged	Iterations	$\|\nabla\|_\infty$	Contraction error
0.594	1.0543	5.23×10^{-3}	yes	3	1.40×10^{-10}	5.74×10^{-11}
0.908	1.0543	5.23×10^{-3}	yes	3	3.33×10^{-11}	5.74×10^{-11}
0.859	1.0543	5.23×10^{-3}	yes	3	4.84×10^{-11}	5.74×10^{-11}
1.426	1.0543	5.23×10^{-3}	yes	3	5.48×10^{-10}	5.74×10^{-11}
0.182	1.0543	5.23×10^{-3}	yes	3	4.47×10^{-11}	5.74×10^{-11}
0.876	**1.0543**	5.23×10^{-3}	**yes**	**3**	4.32×10^{-11}	5.74×10^{-11}
0.254	1.0543	5.23×10^{-3}	yes	3	7.37×10^{-11}	5.74×10^{-11}
1.064	1.0543	5.23×10^{-3}	yes	2	2.18×10^{-9}	5.74×10^{-11}
0.228	1.0543	5.23×10^{-3}	yes	3	6.29×10^{-11}	5.74×10^{-11}
1.168	1.0543	5.23×10^{-3}	yes	3	3.54×10^{-11}	5.74×10^{-11}
0.768	1.0543	5.23×10^{-3}	yes	3	7.80×10^{-11}	5.74×10^{-11}
1.378	1.0543	5.23×10^{-3}	yes	3	3.41×10^{-10}	5.74×10^{-11}
1.088	1.0543	5.23×10^{-3}	yes	2	8.06×10^{-9}	5.74×10^{-11}
0.770	1.0543	5.23×10^{-3}	yes	3	7.83×10^{-11}	5.74×10^{-11}
0.607	1.0543	5.23×10^{-3}	yes	3	1.35×10^{-10}	5.74×10^{-11}
0.444	1.0543	5.23×10^{-3}	yes	3	1.39×10^{-10}	5.74×10^{-11}

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` (per-start values recomputed at each stored σ minimizer, no re-optimization); failed starts from `q8c_blp_only_T100_J3_failed_starts`. Bold: start with the lowest GMM objective across starts..

Table 8: Estimates with and without the Hausman instrument, T=100, J=3.

	True	BLP	BLP + Hausman
Constant	5.000	5.589	4.983***
		(n.a.)	(0.404)
x_2 (uniform)	1.000	1.100	0.967***
		(n.a.)	(0.225)
x_3 (normal)	1.000	1.082	1.037***
		(n.a.)	(0.076)
Price coefficient	−1.000	−1.168	−0.972***
		(n.a.)	(0.116)
σ_α	1.000	1.054	1.007***
		(n.a.)	(0.080)
GMM objective		5.23×10^{-3}	5.56×10^{-3}
Observations (JT)		300	300

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` and `q8c_blp_hausman_T100_J3`; best start of each. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

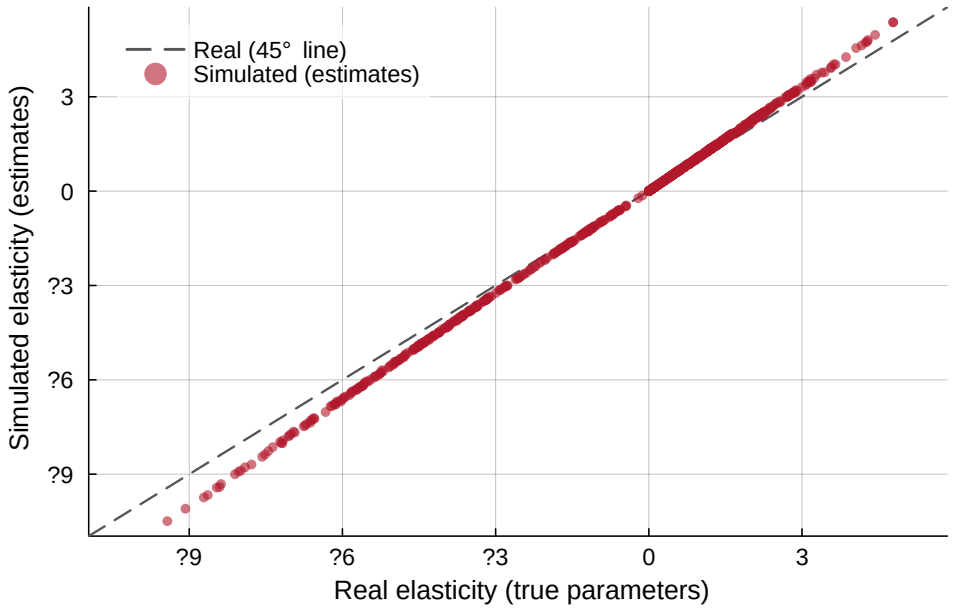
Table 9: First stage of price on the instruments, T=100, J=3 (heteroskedasticity-robust standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; F tests the excluded instruments given x).

Dependent variable: price	BLP	BLP + Hausman
Constant	2.928***	0.145
	(0.251)	(0.423)
x_2	0.550**	0.319
	(0.269)	(0.255)
x_3	0.111	0.057
	(0.074)	(0.072)
BLP rival sum x_2	−0.174	−0.057
	(0.182)	(0.171)
BLP rival sum x_3	−0.131**	−0.104*
	(0.061)	(0.054)
Hausman $\bar{p}_{j,-t}$		0.919***
		(0.126)
R^2	0.037	0.204
F (excluded instruments)	2.9	22.9
Observations	300	300

Source: author's calculations from `100markets3products.mat`; regressions include a constant, x_2 and x_3.

8d

Figure 5: Price elasticities at the estimates versus at the true parameters (all own- and cross-price elements).



Source: author's calculations: elasticities at the estimates of `q8c_blp_only_T100_J3` versus the true-parameter simulation, `100markets3products.mat`.

Table 10: Elasticity of the share of product j with respect to the price of product k: real versus simulated (at the estimates), mean and standard deviation across markets.

	Real		Simulated		Simulated − real	
Element (j,k)	Mean	SD	Mean	SD	Mean	SD
$(1,1)$	−5.097	1.938	−5.608	2.181	−0.511	0.245
$(1,2)$	1.395	1.013	1.549	1.129	0.154	0.117
$(1,3)$	1.000	0.884	1.111	0.984	0.111	0.100
$(2,1)$	0.593	0.696	0.659	0.774	0.065	0.078
$(2,2)$	−2.622	1.666	−2.843	1.845	−0.221	0.181
$(2,3)$	0.985	0.917	1.087	1.014	0.101	0.098
$(3,1)$	0.602	0.700	0.668	0.777	0.066	0.078
$(3,2)$	1.441	1.061	1.592	1.175	0.151	0.116
$(3,3)$	−3.407	1.819	−3.714	2.021	−0.307	0.204

Source: author's calculations: elasticities at the estimates of `q8c_blp_only_T100_J3` versus the true-parameter simulation, `100markets3products.mat`.

8e

Table 11: Estimates by starting point, T=10, J=3, BLP instruments (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

Start σ_0	Constant	x_2	x_3	Price	σ_α	GMM obj.
0.594	**3.223 (2.828)**	**0.428 (0.958)**	**1.155** ** **(0.495)**	**−0.651 (1.025)**	**0.000 (10.635)**	1.00×10^{-1}
0.908	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.859	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
1.426	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.182	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.876	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.254	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
1.064	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.228	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
1.168	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.768	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
1.378	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
1.088	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.770	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.607	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
0.444	3.223 (2.828)	0.428 (0.958)	1.155** (0.495)	−0.651 (1.025)	0.000 (10.635)	1.00×10^{-1}
True	5.000	1.000	1.000	−1.000	1.000	

Source: author's calculations from the stored full estimates `q8e_blp_only_T10_J3` (per-start values recomputed at each stored σ minimizer, no re-optimization). Bold: start with the lowest GMM objective across starts. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

Table 12: Convergence criterion and contraction error by starting point, T=10, J=3.

Start σ_0	$\hat{\sigma}_\alpha$	GMM obj.	Converged	Iterations	$\|\nabla\|_\infty$	Contraction error
0.594	**0.0000**	1.00×10^{-1}	**yes**	**4**	1.49×10^{-11}	1.67×10^{-6}
0.908	0.0000	1.00×10^{-1}	yes	4	4.07×10^{-11}	1.67×10^{-6}
0.859	0.0000	1.00×10^{-1}	yes	4	3.49×10^{-11}	1.67×10^{-6}
1.426	0.0000	1.00×10^{-1}	yes	4	1.80×10^{-10}	1.67×10^{-6}
0.182	0.0000	1.00×10^{-1}	yes	3	1.44×10^{-9}	1.67×10^{-6}
0.876	0.0000	1.00×10^{-1}	yes	4	3.69×10^{-11}	1.67×10^{-6}
0.254	0.0000	1.00×10^{-1}	yes	3	2.91×10^{-9}	1.67×10^{-6}
1.064	0.0000	1.00×10^{-1}	yes	4	6.41×10^{-11}	1.67×10^{-6}
0.228	0.0000	1.00×10^{-1}	yes	3	2.32×10^{-9}	1.67×10^{-6}
1.168	0.0000	1.00×10^{-1}	yes	4	8.61×10^{-11}	1.67×10^{-6}
0.768	0.0000	1.00×10^{-1}	yes	4	2.62×10^{-11}	1.67×10^{-6}
1.378	0.0000	1.00×10^{-1}	yes	4	1.57×10^{-10}	1.67×10^{-6}
1.088	0.0000	1.00×10^{-1}	yes	4	6.86×10^{-11}	1.67×10^{-6}
0.770	0.0000	1.00×10^{-1}	yes	4	2.63×10^{-11}	1.67×10^{-6}
0.607	0.0000	1.00×10^{-1}	yes	4	1.56×10^{-11}	1.67×10^{-6}
0.444	0.0000	1.00×10^{-1}	yes	3	8.81×10^{-9}	1.67×10^{-6}

Source: author's calculations from the stored full estimates `q8e_blp_only_T10_J3` (per-start values recomputed at each stored σ minimizer, no re-optimization); failed starts from `q8e_blp_only_T10_J3_failed_starts`. Bold: start with the lowest GMM objective across starts..

Table 13: Estimates with and without the Hausman instrument, T=10, J=3.

	True	BLP	BLP + Hausman
Constant	5.000	3.223	4.092***
		(2.828)	(1.277)
x_2 (uniform)	1.000	0.428	0.496
		(0.958)	(0.901)
x_3 (normal)	1.000	1.155**	1.009***
		(0.495)	(0.308)
Price coefficient	−1.000	−0.651	−0.996**
		(1.025)	(0.482)
σ_α	1.000	0.000	0.001
		(10.635)	(9.576)
GMM objective		1.00×10^{-1}	5.50×10^{-1}
Observations (JT)		30	30

Source: author's calculations from the stored full estimates `q8e_blp_only_T10_J3` and `q8e_blp_hausman_T10_J3`; best start of each. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

8f

Table 14: Estimates imposing $E[\xi_{jt} p_{jt}] = 0$ (price as its own instrument) versus the original IV specification.

	True	BLP (IV)	Price as instrument
Constant	5.000	5.589	4.775***
		(n.a.)	(0.352)
x_2 (uniform)	1.000	1.100	0.922***
		(n.a.)	(0.216)
x_3 (normal)	1.000	1.082	1.022***
		(n.a.)	(0.078)
Price coefficient	−1.000	−1.168	−0.904***
		(n.a.)	(0.083)
σ_α	1.000	1.054	0.989***
		(n.a.)	(0.085)
GMM objective		5.23×10^{-3}	5.83×10^{-3}
Observations (JT)		300	300

Source: author's calculations from the stored full estimates `q8c_blp_only_T100_J3` (original) and `q8f_blp_price_T100_J3`.

Table 15: Price-as-instrument estimates with and without the Hausman instrument.

	True	Price as instrument	Price + Hausman
Constant	5.000	4.775***	4.867***
		(0.352)	(0.341)
x_2 (uniform)	1.000	0.922***	0.945***
		(0.216)	(0.220)
x_3 (normal)	1.000	1.022***	1.036***
		(0.078)	(0.076)
Price coefficient	−1.000	−0.904***	−0.925***
		(0.083)	(0.082)
σ_α	1.000	0.989***	1.015***
		(0.085)	(0.080)
GMM objective		5.83×10^{-3}	3.69×10^{-1}
Observations (JT)		300	300

Source: author's calculations from the stored full estimates `q8f_blp_price_T100_J3` and `q8f_blp_price_hausman_T100_J3`.

Question 9: Cost shifters and the supply side

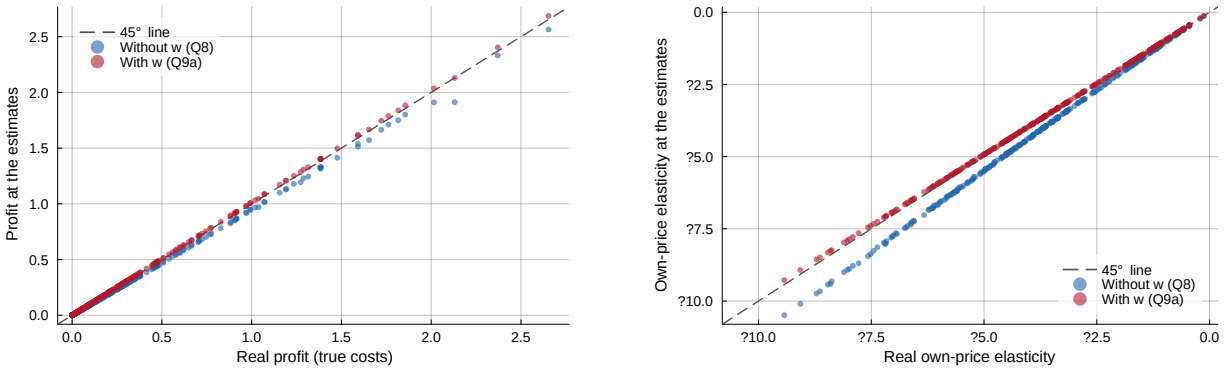
9a

Table 16: Demand estimates with and without the cost shifter w_j in the instrument set (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	True	BLP (Q8)	BLP $+ w$ (Q9a)	BLP + Haus. (Q8)	BLP + Haus. $+ w$ (Q9a)
Constant	5.000	5.589	5.012***	4.983***	5.005***
		(n.a.)	(0.396)	(0.404)	(0.397)
x_2 (uniform)	1.000	1.100	0.973***	0.967***	0.971***
		(n.a.)	(0.227)	(0.225)	(0.226)
x_3 (normal)	1.000	1.082	1.039***	1.037***	1.037***
		(n.a.)	(0.076)	(0.076)	(0.076)
Price coefficient	−1.000	−1.168	−0.981***	−0.972***	−0.980***
		(n.a.)	(0.113)	(0.116)	(0.113)
σ_α	1.000	1.054	1.009***	1.007***	1.005***
		(n.a.)	(0.080)	(0.080)	(0.080)
GMM objective		5.23×10^{-3}	5.52×10^{-3}	5.56×10^{-3}	2.00×10^{-1}
Observations (JT)		300	300	300	300

Source: author's calculations from the stored full estimates `q8c_*` and `q9a_*` (T=100, J=3).

Figure 6: Profits and own-price elasticities at the estimates, with and without the cost shifter, against real values.



(a) Profits

(b) Own-price elasticities

Source: author's calculations from the stored full estimates `q8c_*` and `q9a_*` (T=100, J=3); profits $(p - \hat{c})s$ with costs recovered under Bertrand pricing; author's calculations from the stored full estimates `q8c_*` and `q9a_*` (T=100, J=3).

9b

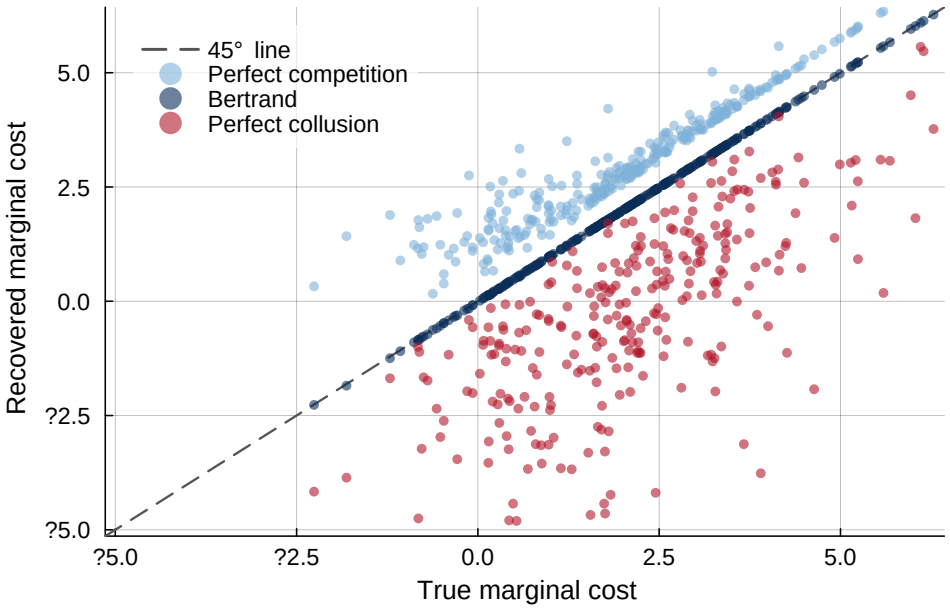
Table 17: Small-sample estimates (T=10, J=3) with and without the cost shifter w_j (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	True	BLP (Q8e)	BLP $+w$ (Q9b)	BLP + Haus. (Q8e)	BLP + Haus. $+w$ (Q9b)
Constant	5.000	3.223	1.878	4.092***	3.103**
		(2.828)	(3.068)	(1.277)	(1.545)
x_2 (uniform)	1.000	0.428	0.350	0.496	0.417
		(0.958)	(1.352)	(0.901)	(0.985)
x_3 (normal)	1.000	1.155**	1.498**	1.009***	1.184***
		(0.495)	(0.626)	(0.308)	(0.357)
Price coefficient	−1.000	−0.651	0.069	−0.996**	−0.591
		(1.025)	(1.139)	(0.482)	(0.387)
σ_α	1.000	0.000	0.605	0.001	0.138
		(10.635)	(1.484)	(9.576)	(1.436)
GMM objective		1.00×10^{-1}	5.51×10^{0}	5.50×10^{-1}	6.73×10^{0}
Observations (JT)		30	30	30	30

Source: author's calculations from the stored full estimates `q8e_*` and `q9b_*` (T=10, J=3).

9d

Figure 7: True versus recovered marginal costs under perfect competition, Bertrand pricing and perfect collusion (axes cover the 1st to 99th percentile of the plotted values).



Source: author's calculations: cost inversion at the demand estimates of `q9a_blp_hausman_w_T100_J3`.

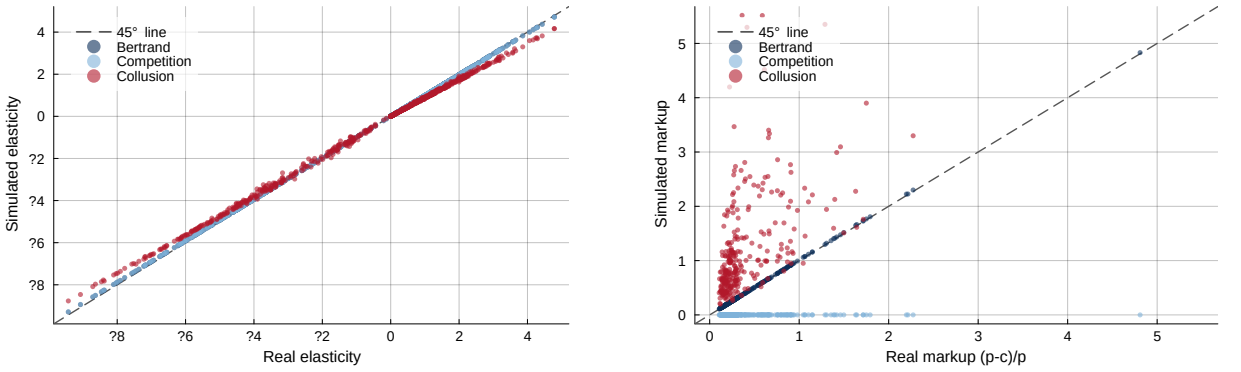
9e

Table 18: Joint demand and supply estimates under three conduct assumptions (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	True	Bertrand	Competition	Collusion
Constant	5.000	5.019***	5.005***	4.109***
		(0.397)	(0.397)	(0.966)
x_2 (uniform)	1.000	0.975***	0.971***	0.778***
		(0.226)	(0.226)	(0.261)
x_3 (normal)	1.000	1.039***	1.037***	0.895***
		(0.076)	(0.076)	(0.137)
Price coefficient	−1.000	−0.983***	−0.980***	−0.781***
		(0.113)	(0.113)	(0.251)
γ_0 (cost constant)	2.000	1.940***	2.951***	−1.152***
		(0.074)	(0.055)	(0.437)
γ_z (z_{jt})	1.000	0.916***	0.790***	0.598
		(0.056)	(0.051)	(0.555)
γ_w (w_j)	1.000	0.979***	0.776***	0.622
		(0.075)	(0.069)	(0.677)
σ_α	1.000	1.009***	1.005***	0.701**
		(0.079)	(0.080)	(0.312)
GMM objective		3.15×10^0	8.09×10^0	2.86×10^1
Observations (JT)		300	300	300

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Figure 8: Real versus simulated elasticities and markups under the three conduct assumptions.



(a) Elasticities

(b) Markups (axes cover the 1st to 99th percentile)

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Table 19: Descriptive statistics of real and simulated elasticities.

	Mean	SD	Median	Q25	Q75
Own-price elasticities					
Real	−3.709	2.080	−3.650	−5.084	−1.744
Bertrand	−3.661	2.049	−3.613	−5.012	−1.722
Competition	−3.658	2.047	−3.609	−5.008	−1.722
Collusion	−3.516	1.930	−3.448	−4.844	−1.705
Cross-price elasticities					
Real	1.003	0.947	0.771	0.210	1.475
Bertrand	0.992	0.937	0.761	0.207	1.459
Competition	0.990	0.935	0.760	0.207	1.456
Collusion	0.876	0.822	0.676	0.184	1.303

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Table 20: Descriptive statistics of real and simulated markups.

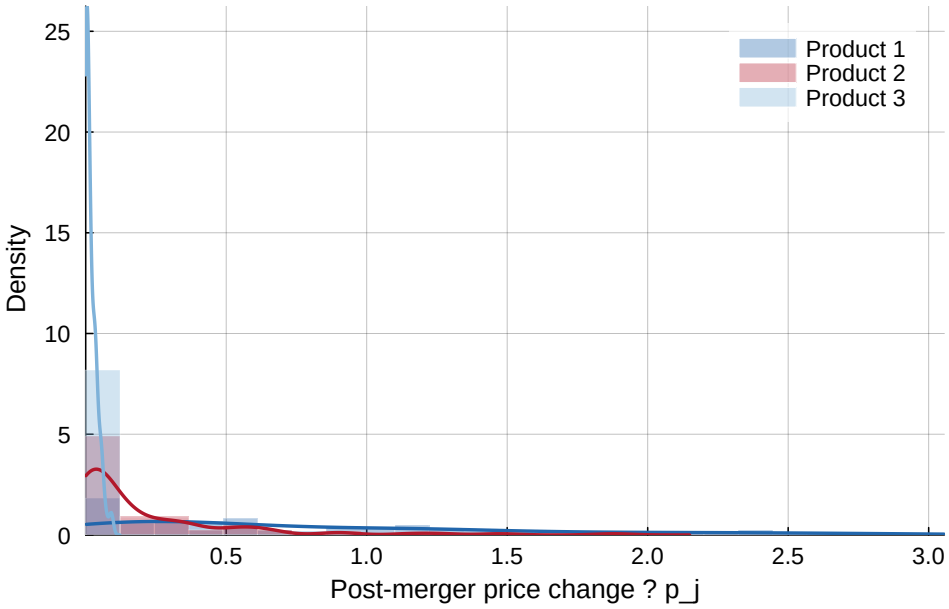
	Mean	SD	Median	Q25	Q75
Real	0.472	0.634	0.274	0.197	0.574
Bertrand	0.477	0.636	0.277	0.200	0.581
Competition	0.000	0.000	0.000	0.000	0.000
Collusion	4.275	35.432	0.988	0.629	1.694

Source: author's calculations from the stored full estimates `q9e_system_*_T100_J3`.

Question 10: Merger simulation

10b

Figure 9: Distribution and kernel density of the post-merger price changes of products 1, 2 (merging) and 3 (non-merging) across markets.



Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`.

Table 21: Descriptive statistics of post-merger price changes by product.

Product	Mean	SD	Median	Q25	Q75
1	0.800	0.771	0.539	0.170	1.204
2	0.213	0.329	0.075	0.017	0.270
3	0.019	0.021	0.012	0.002	0.032

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`.

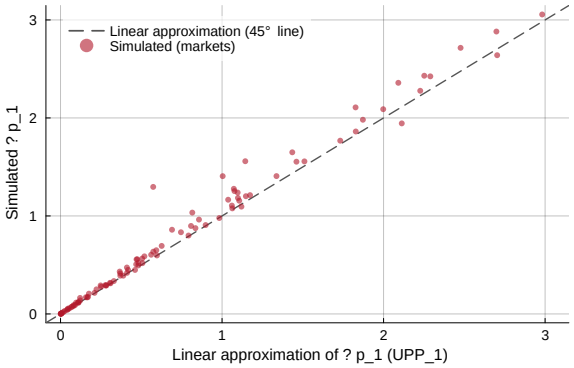
Table 22: Descriptive statistics of the post-merger changes in consumer, producer and total surplus across markets.

	Mean	SD	Median	Q25	Q75
Δ Consumer surplus	−0.084	0.147	−0.029	−0.092	−0.008
Δ Producer surplus	0.045	0.078	0.017	0.004	0.048
Δ Total surplus	−0.039	0.074	−0.009	−0.043	−0.002

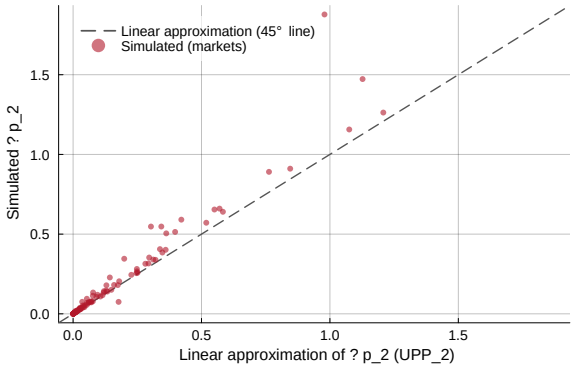
Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`.

10c

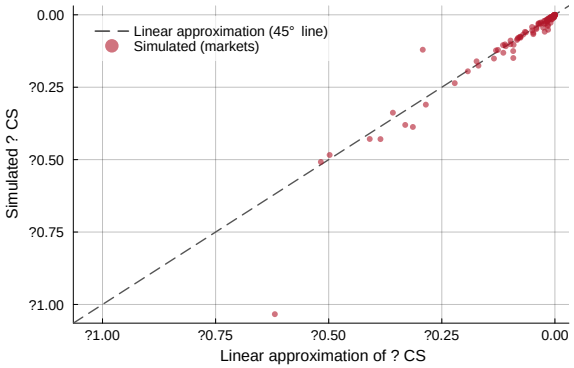
Figure 10: Simulated merger effects against their linear (upward pricing pressure) approximations, one point per market.



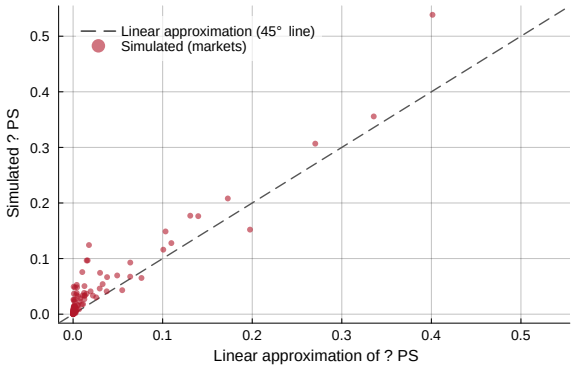
(a) Price change of product 1



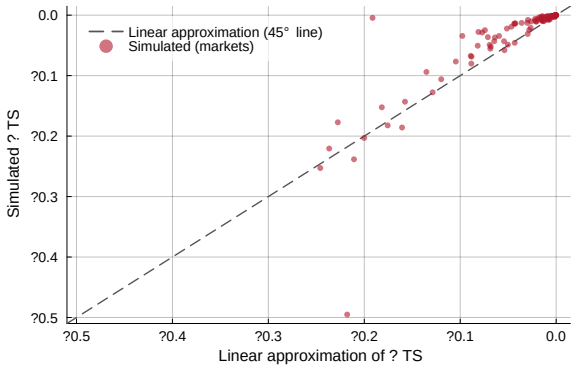
(b) Price change of product 2



(c) Change in consumer surplus



(d) Change in producer surplus



(e) Change in total surplus

Source: author's calculations: merger counterfactual (firms 1 and 2, no cost efficiencies) at the estimates of `q9e_system_bertrand_T100_J3`; diversion ratios and $\text{UPP}_j = D_{jk}(p_k - c_k)$ from the pre-merger estimates.

Question 11: Newer instruments and substitution diagnostics

11a

Table 23: Price first stage with each instrument family, T=100, J=5 (heteroskedasticity-robust standard errors in parentheses; F tests the excluded instruments given x).

Dependent variable: price	Hausman	BLP	GH local	GH quadratic	All
Constant	0.124	2.030***	2.258***	1.805***	0.093
	(0.211)	(0.292)	(0.205)	(0.180)	(0.428)
x_2	−0.091	−0.217	−0.168	−0.175	−0.068
	(0.205)	(0.226)	(0.231)	(0.229)	(0.210)
x_3	−0.009	−0.061	−0.069	−0.067	−0.019
	(0.057)	(0.063)	(0.064)	(0.063)	(0.058)
Hausman $\bar{p}_{j,-t}$	0.959***				0.958***
	(0.088)				(0.090)
BLP rival sum x_2		−0.012			−0.040
		(0.127)			(0.114)
BLP rival sum x_3		0.013			−0.001
		(0.031)			(0.029)
GH local rival x_2			−0.044		0.077
			(0.061)		(0.077)
GH local rival x_3			−0.089		−0.086
			(0.058)		(0.077)
GH quadratic rival x_2				0.190	0.226
				(0.136)	(0.178)
GH quadratic rival x_3				0.008	−0.001
				(0.009)	(0.013)
R^2	0.200	0.004	0.009	0.009	0.207
F (excluded instruments)	121.7	0.1	1.3	1.3	17.9
Observations	500	500	500	500	500

Source: author's calculations from `100markets5products.mat`; first-stage regressions include a constant, x_2 and x_3.

Table 24: Estimates and final GMM objective by instrument set, T=100, J=5 (standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	True	BLP	GH	GH + Hausman
Constant	5.000	3.870	5.519***	5.295***
		(1.25×10^4)	(1.355)	(0.641)
x_2 (uniform)	1.000	0.861	0.691*	0.706***
		(1390.753)	(0.373)	(0.173)
x_3 (normal)	1.000	1.039	0.850***	1.007***
		(523.764)	(0.097)	(0.090)
Price coefficient	−1.000	−0.418	−1.552**	−1.100***
		(6692.320)	(0.719)	(0.076)
σ_α	1.000	0.927	0.000	0.950***
		(377.796)	(17.433)	(0.208)
GMM objective		6.28×10^{-1}	2.54×10^{-1}	3.56×10^{0}
Observations (JT)		500	500	500

Source: author's calculations from the stored full estimates `q11a_*_T100_J5`; best start of each. No fixed effects are used in any specification. (n.a.): estimated variance not positive; very large standard errors indicate weak identification..

Table 25: Sensitivity to starting values, T=100, J=5.

Instruments	Starts ok	Starts failed	Converged	Min $\hat{\sigma}_\alpha$	Max $\hat{\sigma}_\alpha$	Min obj.	Max obj.	Mean iter.
BLP	7	9	0	0.8484	0.9922	6.28×10^{-1}	6.89×10^{-1}	1.0
GH	16	0	0	0.0000	0.3918	2.54×10^{-1}	4.70×10^{-1}	1.1
GH + Hausman	6	10	6	0.9504	0.9504	3.56×10^{0}	3.56×10^{0}	3.5

Source: author's calculations from the stored full estimates `q11a_*_T100_J5` and the corresponding `*_failed_starts` files; summary over all starting points.

11b

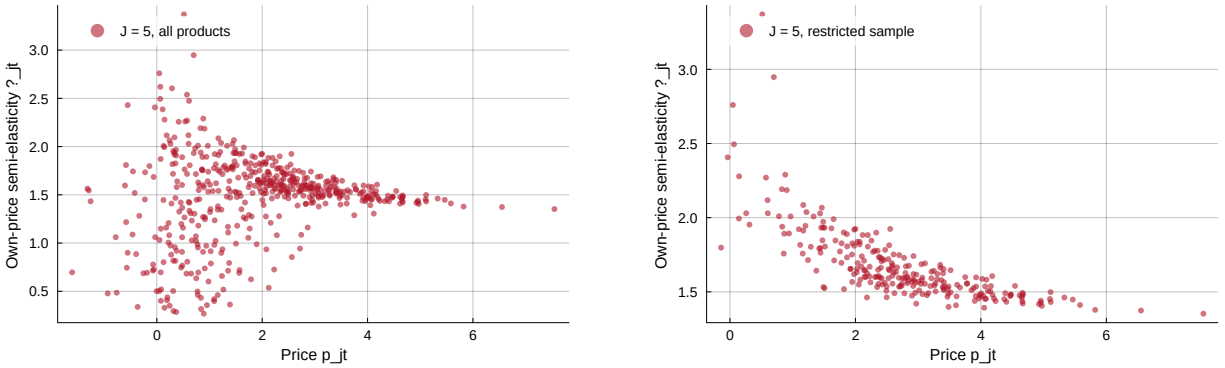
Table 26: Variation of the price and of each instrument explained by market and product fixed effects, T=100, J=5; the last two columns show what remains after both sets of effects.

Variable	SD	R^2 market FE	R^2 product FE	R^2 both	SD after both	Share of variance left
Price p_{jt}	1.435	0.196	0.215	0.411	1.101	0.589
Hausman $\bar{p}_{j,-t}$	0.666	0.000	1.000	1.000	0.011	0.000
BLP rival sum x_2	0.555	0.796	0.001	0.797	0.250	0.203
BLP rival sum x_3	2.076	0.826	0.003	0.830	0.857	0.170
GH local rival x_2	1.086	0.426	0.011	0.437	0.815	0.563
GH local rival x_3	1.097	0.397	0.008	0.405	0.846	0.595
GH quadratic rival x_2	0.474	0.565	0.009	0.574	0.309	0.426
GH quadratic rival x_3	6.528	0.538	0.009	0.547	4.393	0.453

Source: author's calculations from `100markets5products.mat`; R^2 from regressing each variable on market dummies, product dummies, or both (balanced panel).

11c

Figure 11: Own-price semi-elasticity against price.



(a) All product-markets

(b) Restricted sample ($|i_{jt}^{scp}/\eta_{jt}| < 0.25$)

Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; $\eta_{jt} = -s_{jt}^{-1}\partial s_{jt}/\partial p_{jt}$; author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; $\eta_{jt} = -s_{jt}^{-1}\partial s_{jt}/\partial p_{jt}$; approximation-error ratio as in `pset_io`.

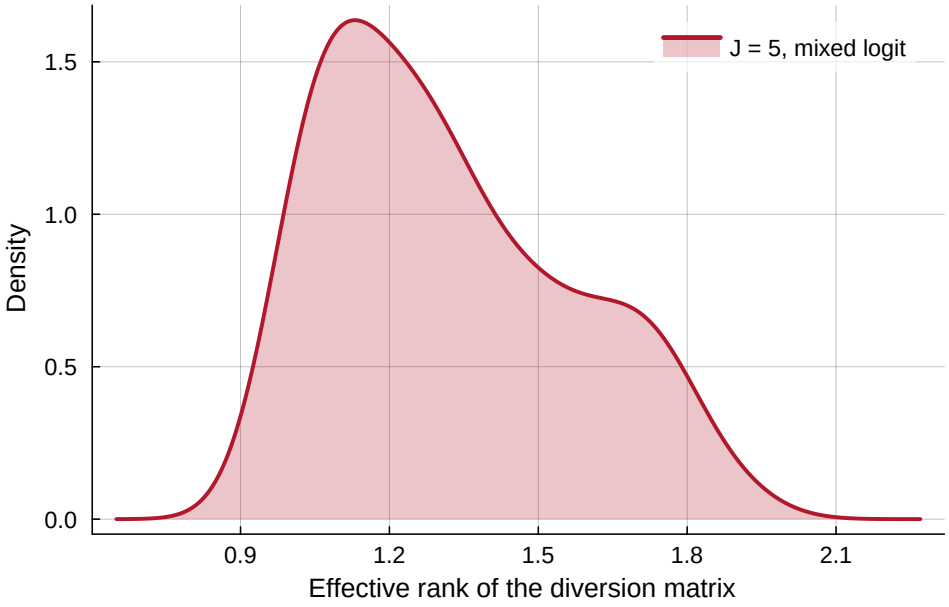
Table 27: Flexible regression of the own-price semi-elasticity on price and characteristics (the same specification in every column). Columns differ only in the sample: the cut-off on $|i^{scp}_{jt}/\eta_{jt}|$ falls from left to right, so the sample becomes more restricted (heteroskedasticity-robust standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

Cut-off on $\lvert i^{scp}_{jt}/\eta_{jt}\rvert$	Full	2.00	1.00	0.75	0.50	0.25	0.10
Constant	1.603***	1.885***	2.037***	2.081***	2.236***	2.381***	2.294***
	(0.117)	(0.106)	(0.096)	(0.093)	(0.103)	(0.097)	(0.129)
p	0.103**	−0.078*	−0.163***	−0.192***	−0.274***	−0.339***	−0.295***
	(0.046)	(0.045)	(0.042)	(0.042)	(0.048)	(0.044)	(0.061)
x_2	−0.858***	−0.417*	−0.291	−0.272	−0.341*	−0.358*	−0.363
	(0.295)	(0.225)	(0.190)	(0.185)	(0.187)	(0.209)	(0.238)
x_3	−0.227***	−0.191***	−0.132***	−0.131***	−0.131***	−0.088*	−0.086
	(0.039)	(0.034)	(0.038)	(0.040)	(0.044)	(0.047)	(0.055)
p^2	−0.025***	−0.002	0.007	0.012**	0.023***	0.028***	0.024***
	(0.006)	(0.005)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)
$1/p$	0.002***	0.009	0.014	0.013	0.012	0.005	0.020**
	(0.000)	(0.014)	(0.014)	(0.012)	(0.010)	(0.007)	(0.009)
x_2^2	0.561**	0.248	0.093	0.114	0.193	0.100	0.143
	(0.231)	(0.154)	(0.125)	(0.121)	(0.119)	(0.134)	(0.155)
x_3^2	−0.007	−0.010	−0.011	−0.003	−0.008	−0.011	−0.007
	(0.013)	(0.011)	(0.011)	(0.014)	(0.013)	(0.013)	(0.013)
$p \cdot x_2$	0.081*	0.044	0.056	0.041	0.040	0.074**	0.056*
	(0.047)	(0.041)	(0.039)	(0.039)	(0.033)	(0.033)	(0.033)
$p \cdot x_3$	0.052***	0.044***	0.029**	0.033**	0.032**	0.019	0.019
	(0.013)	(0.011)	(0.012)	(0.013)	(0.014)	(0.015)	(0.017)
R^2	0.186	0.312	0.481	0.542	0.651	0.689	0.662
Observations	500	414	368	347	312	254	194

Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; $\eta_{jt} = -s_{jt}^{-1}\partial s_{jt}/\partial p_{jt}$; each column keeps the product-markets with $|i^{scp}_{jt}/\eta_{jt}|$ below the cut-off in the header.

11d

Figure 12: Density of the effective rank (exponential of the entropy of normalized singular values) of the model-implied diversion matrix across markets.



Source: author's calculations at the stored estimates `q11a_gh_hausman_T100_J5`; mixed logit only.